



Helping New Zealanders Build & Modify Safe Vehicles

LVV Operating Requirements Schedule

Chapter 3 LVV Certification Categories

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Approval Record

Signed in accordance with clause 1.3(5) of the Low Volume Vehicle Code of LVVTA, on by:			
New Zealand Transport Agency		Low Volume Vehicle Technical Association	
Name	Signature	Name	Signature
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Amendment Record

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Note 1 The first ten amendment processes to the LVV ORS (Amendment #s 1-10), carried out between August 2003 and June 2017, were made to the complete ORS document. From Amendment # 11 (which is Version 12, issued 1 October 2024), amendments are carried out to individual chapters. Note 2 Text highlighted in grey shows amendments that have been made subsequent to the previous version of this chapter, and a grey vertical stroke to the left of the text denotes important new or changed information (which may include information which has been removed).			

About the LVV Operating Requirements Schedule

The LVV Operating Requirements Schedule (LVV ORS), and its sub-set of LVV ORS chapters (the chapters) set out the operational systems and processes which enables the LVV certification system to function effectively. Whereas the *Low Volume Vehicle Code* provides the legal platform upon which the LVV certification system operates, the LVV ORS provides robust operational systems and processes to ensure that LVV certification outcomes are consistent, fair, transparent, and of a high quality.

Author, Publisher, & Owner

This chapter is authored, published, and owned by the Low Volume Vehicle Technical Association Incorporated (LVVTA). LVVTA is an incorporated society established in 1992, that represents a group of specialist automotive organisations (in turn representing approximately 150,000 members) who are dedicated to ensuring that vehicles, when scratch-built or modified, meet the highest practicable safety standards.

The information in this chapter has stemmed from work undertaken by LVVTA founding member organisations that commenced in 1989 and has been progressively developed as an integral part of the New Zealand Government's land transport regulatory system, by agreement and in consultation with the New Zealand Transport Agency (NZTA).

As a result, the considerable experience in specialist certification management built up by LVVTA and the specialist automotive member groups over the past several decades can be of benefit to members of the New Zealand public who also wish to build or modify motor vehicles.

LVVTA's contact details are:

- Phone: +(00) 64 4 238 4343
- Email: info@lvvta.org.nz
- Postal address: P. O. Box 50-600, Porirua 5240, Wellington, New Zealand
- Website: www.lvvta.org.nz

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Therefore, this and all other LVVTA documents should not be relied upon without first ensuring that the version number (on the right-hand side of the header above) is the current version – please visit the LVV ORS area of www.lvvta.org.nz to check that this chapter is in fact the latest version.

User's Feedback

This chapter is constantly undergoing an evolutionary development process in order to keep pace with changing trends and technology. To assist in this, LVVTA invites users of the chapter to engage in an ongoing consultation process with us by making submissions for any changes, additions, or clarifications which might improve the chapter, at any time.

Any submissions made via this rolling consultation process will be thoroughly considered, and incorporated, where appropriate, at the next available amendment opportunity.

Submissions should be made to submission@lvvta.org.nz with the name of this chapter in the Subject line.

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Chapter 3

LVV Certification Categories

Purpose of this Chapter

The purpose of this LVV Operating Requirements Schedule chapter (this chapter) is to explain the various LVV certification categories for which an LVV Certifier may be appointed to certify, and the technical limitations (by either modification or vehicle type) that apply to each of the different categories.

The LVV certification category structure is tiered, in recognition that very few (if any) LVV Certifiers will possess the necessary background to competently assess every type of motor vehicle modification and construction feature, and vehicle type, within the LVV certification environment. This tiered structure has been established to ensure all LVV Certifiers are operating within their field of knowledge and experience.

This chapter should be read in conjunction with *LVV ORS Chapter 4: LVV Certifier Background Criteria* (which provides details of the knowledge and practical experience required in order to be appointed for each of the LVV certification categories), and *LVV ORS Chapter 5: LVV Certifier Application & Appointment* (which provides details of the various processes involved in becoming appointed as an LVV Certifier).

Italics are used throughout this chapter when referencing 'external documents' that are not part of this chapter.

Section 1 LVV Certification Category Background Information

1.1 Introduction

Because of the complexity and diversity of the vehicle modification and construction industry, LVVTA has established a range of LVV certification categories that enable an LVV Certifier's particular area of experience to be appropriately matched to specific LVV certification activities. This chapter lists the range of LVV certification categories for which an LVV Certifier may be appointed by the New Zealand Transport Agency (NZTA), together with the technical limitations of each of those categories.

1.2 Explanation of vehicle 'classes' referred to

The content of this chapter refers to specific 'types' of vehicles (as defined in NZTA's 'Table A vehicle classes'). The classes of vehicles applicable to LVV certification are defined as (in simplest terms):

- Class M-group: normal passenger-carrying cars, station wagons, and vans.
- Class NA: Light goods-carrying utilities and vans.
- Class L-group: Motorcycles.
- Class LE: Motor-tricycles (Trikes).

A full list and descriptions of NZTA's Table A vehicle classes are provided in the *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>.

1.3 General requirements

- 1.3(1) An LVV Certifier must LVV certify only those modifications and construction features relevant to the LVV certification category(s) for which they are appointed, as specified in section 2 to section 6.

- 1.3(2) The list of modifications and construction features which **may be certified**, and which **must not be certified**, by an LVV Certifier with a particular category, are specified in section 2 to section 6 of this chapter, which must be read together before determining the outcome on any given modification subject.
- 1.3(3) Where a modification or construction feature present on a low volume vehicle undergoing LVV certification is not listed, LVVTA will make a determination as to which LVV certification category is required to LVV certify the vehicle (see Note 1 below).

Note 1 LVVTA is in the unique position of continually overseeing all light vehicle modification work in New Zealand on an every-day basis, and is therefore able to make sound and consistent judgements relating to aligning an LVV Certifier's knowledge and practical experience with modification and vehicle types. In turn, this experience within LVVTA ensures consistency of decisions across all LVV Certifiers.

- 1.3(4) A previously LVV certified low volume vehicle being re-certified for a subsequent modification must be certified by an LVV Certifier who is authorised for the LVV certification category that would have been required for the original certification of the vehicle (see Note 1 below).

Note 1 This requirement is necessary to achieve the correct safety outcomes. For example, a scratch-built sports car (originally requiring LVV certification category LV1D) with a subsequent minor suspension modification, must be LVV certified by an LV1D Certifier, even though, if that same modification was performed on a modified production vehicle, that vehicle could be LVV certified by an LV1A Certifier.

This is because a comprehensive knowledge of the vehicle, and type of modifications in question, is required in order to identify and understand any potential flow-on influences of any subsequent modifications on inter-related aspects of the original vehicle's design.

Section 2 Category 1: Passenger & Light Goods Vehicles

2.1 Category LV1A Modified Production – Limited

- 2.1(1) An LVV Certifier appointed to certify under LVV certification category 'LV1A Modified Production - Limited' (an LV1A Certifier) is authorised to certify all Class M-group and Class NA modified production low volume vehicles to the extent specified in 2.1 as **'may be certified'** (see Note 1 below).

Note 1 Class M-group means passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

Engine and drive-train

- 2.1(2) Engine and drive-train modifications and conversions which **may be certified** by an LV1A Certifier **are** those where:
- (a) the engine is in its original location; and
 - (b) no significant structural modifications have been carried out to the firewall, floor, driveshaft or gearbox tunnel (see Note 1 below); and
 - (c) no significant structural modifications have been carried out to the chassis, sub-frame, or suspension cross-members (see Note 2 below).

Note 1 These modifications are particularly relevant in the case of a unitary-constructed vehicle.

Note 2

A 'significant structural modification' does not include:

- the fabrication of typical mounts or mounting systems to attach an engine, gearbox, or differential; or
- the substitution of a 'bolt-in' cross-member from the same vehicle make and model variant, or aftermarket equivalent, provided the vehicle's OEM steering and suspension geometry is maintained; or
- minor trimming of the flanges of a chassis, sub-frame, or cross-member, provided that any strength removed by the trimming has been reinstated.

2.1(3) Engine and drive-train modifications and conversions which **must not be certified** by an LV1A Certifier are those which are not listed in 2.1(2).

Braking systems

2.1(4) Braking system modifications and conversions which **may be certified** by an LV1A Certifier are, subject to 2.1(5):

- substitution and adaptation of 'bolt-on' OEM or aftermarket braking components or systems; and
- conversion of mechanical braking systems to hydraulic braking systems which use 'bolt-on' OEM components or systems; and
- relocation and remounting of a master cylinder; and
- conversion to an adjustable-bias twin single-circuit master cylinder system; and
- disabling or removal of OEM anti-lock braking systems; and
- 'bolt-on' OEM-type stub axle or hub conversions which change the pitch circle diameter of the hub.

2.1(5) Braking system modifications and conversions which **must not be certified** by an LV1A Certifier are:

- modifications to stub axle housings or spindles; or
- significant modifications to an OE brake pedal (see Note 1 below); or
- aftermarket brake pedals; or
- custom-built braking components or systems; or
- those which are not listed in 2.1(4).

Note 1

The only modification to a brake pedal which would not be considered a 'significant modification', is the attachment of a bracket to accommodate a brake light switch, provided that the attachment of the bracket could not, in any way, affect the strength of the brake pedal.

Steering systems

2.1(6) Steering system modifications and conversions which **may be certified** by an LV1A Certifier are, subject to 2.1(7):

- substitution and adaptation of 'bolt-on' OEM or aftermarket steering components or systems; and
- conversion of manual steering systems to power steering systems, and vice-versa, which use 'bolt-on' OEM components; and

- (c) aftermarket 'bolt-on' HICAS lock-bars, specifically limited to the Nissan HICAS system only (see Note 1 below); and
- (d) right-hand drive conversions, provided that:
 - (i) the vehicle was manufactured before 1980; and
 - (ii) the conversion incorporates OEM components; and
 - (iii) the conversion is of a direct 'mirror-image' nature; and
 - (iv) the vehicle manufacturer made provision during the manufacturing process for the vehicle to be assembled in either left-hand or right-hand drive form;
- and
- (e) modifications directly associated with a body-lift, which are limited to:
 - (i) the adjustment of the positioning of the splined section of the OEM steering intermediate shaft; or
 - (ii) a 'bolt-on' aftermarket component (which does not incorporate welding and that is catalogued for the make and model of vehicle) which is designed to lengthen the OEM steering intermediate shaft; or
 - (iii) the repositioning of a steering column support bracket.

Note 1 The HICAS lock-bar system is limited to aftermarket lock-bars fitted to disable the Nissan HICAS system, which feature welded mounting tabs that wrap around part of the bar's circumference. Lock-bars with mounting tabs that only butt-weld to the lock-bar must not be LVV certified by an LV1A Certifier.

2.1(7) Steering system modifications and conversions which **must not be certified** by an LV1A Certifier are:

- (a) with the exception of those listed in 2.1(6)(c) and (e), any modifications and conversions which change, substitute, or reposition any geometry-critical steering components or systems; or
- (b) modifications to any individual steering component or mounting system; or
- (c) aftermarket steering boxes or steering racks; or
- (d) custom-built steering components or systems; or
- (e) those which are not listed in 2.1(6).

Suspension systems

2.1(8) Suspension system modifications and conversions which **may be certified** by an LV1A Certifier are, subject to 2.1(9):

- (a) substitution and adaptation of 'bolt-on' OEM or aftermarket suspension components or systems; and
- (b) complete 'bolt-on' OEM suspension airbag to OEM or aftermarket coil spring conversions, provided that no change to the suspension's load-path or geometry has occurred; and
- (c) complete 'bolt-on' OEM coil spring to OEM airbag conversions, provided that no change to the suspension's load-path or geometry has occurred; and

- (d) repositioning of a non-load bearing shock absorber (see Note 1 below); and
- (e) 'bolt-on' aftermarket (loaded or unloaded) suspension arms; and
- (f) 'bolt-on' aftermarket (dropped spindle-type) stub axle conversions; and
- (g) 'bolt-on' OEM-type stub axle conversions that change the hub pitch circle diameter; and
- (h) 'bolt-on' aftermarket roll-centre adjusters; and
- (i) 'bolt-on' aftermarket drop-boxes (fitted for the purpose of correcting caster geometry).

Note 1 A 'non-load bearing shock absorber' means a shock absorber that operates independently of a spring or a bump-stop.

- 2.1(9) Suspension system modifications and conversions which **must not be certified** by an LV1A Certifier are:
- (a) changes or modifications which could affect the relationship between the steering geometry and the suspension geometry; or
 - (b) changes to the suspension configuration (see Note 1 below); or
 - (c) changes or modifications which could affect the suspension load-path; or
 - (d) custom-built suspension components or systems; or
 - (e) complete suspension assembly changes; or
 - (f) structural modifications to the suspension cross-members; or
 - (g) airbag or hydraulic suspension conversions, other than those specified in 2.1(8)(c); or
 - (h) those which are not listed in 2.1(8).

Note 1 An example of a change to the suspension configuration is a change from an OEM double wishbone system to a Macpherson strut system.

Wheels and tyres

- 2.1(10) All wheel and tyre modifications and changes **may be certified** by an LV1A Certifier.

Fuel systems

- 2.1(11) All fuel system modifications and conversions **may be certified** by an LV1A Certifier, including custom-built fuel tanks and systems.

Glazing and vision

- 2.1(12) All glazing and vision modifications and conversions **may be certified** by an LV1A Certifier, including windscreen wipe and wash systems.

Body modifications

- 2.1(13) Body modifications and conversions which **may be certified** by an LV1A Certifier are, subject to 2.1(14):
- (a) non-structural body modifications such as body-kits, spoilers, substituted or modified door opening systems, fuel filling systems, and intercooler installations; and
 - (b) a body-lift of up to 50mm on a four-wheel drive vehicle, including associated steering modifications limited to those specified in 2.1(6)(e); and

- (c) modifications relating to external projection and field of vision; and
- (d) lighting modifications and conversions.

2.1(14) Body modifications and conversions which **must not be certified** by an LV1A Certifier are:

- (a) structural body modifications and conversions; or
- (b) substantial or complete body substitution; or
- (c) substantial or complete custom body construction; or
- (d) those which are not listed in 2.1(13).

Interior

2.1(15) Interior modifications and conversions which **may be certified** by an LV1A Certifier are, subject to 2.1(16):

- (a) non-structural interior modifications and conversions; and
- (b) unstressed 'bolt-in' seating substitution.

2.1(16) Interior modifications and conversions which **must not be certified** by an LV1A Certifier are:

- (a) seatbelt anchorage installations; or
- (b) seat modifications; or
- (c) stressed seat changes or installations; or
- (d) those which are not listed in 2.1(15).

2.2 Category LV1B Modified Production – Extended

2.2(1) An LVV Certifier appointed to certify under LVV certification category 'LV1B Modified Production – Extended' (an LV1B Certifier) is authorised to certify all Class M-group and Class NA modified production low volume vehicles to the extent specified in 2.2 as **'may be certified'** (see Note 1 below).

Note 1 **Class M-group means** passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

Seatbelt anchorages

2.2(2) All seatbelt anchorage installations **may be certified** by an LV1B Certifier.

Seat design and anchorages

2.2(3) All seat design, modification, construction, and installations, including stressed seats, **may be certified** by an LV1B Certifier.

Adaptive control systems

2.2(4) Adaptive control systems to enable a disabled driver to drive a motor vehicle which **may be certified** by an LV1B Certifier comprise, subject to 2.2(5):

- (a) mechanical adaptive hand controls; and
- (b) modifications to foot controls; and

- (c) power steering modifications which meet *LVV Information Sheet # 02-2007 - Power Steering Torsion Bar Modifications* (see Note 1 below).

Note 1 *LVV Information Sheet # 02-2007: Power Steering Torsion Bar Modifications* is available to the public electronically, free of charge, from the LVVTA website www.lvvta.org.nz

- 2.2(5) Adaptive control systems to enable a disabled driver to drive a motor vehicle which **must not be certified** by an LV1B Certifier are:

- (a) electro-mechanical driver control systems; or
- (b) those which are not listed in 2.2(4).

Frontal impact systems

- 2.2(6) All modifications to, and which may affect, a frontal impact restraint system, **may be certified** by an LV1B Certifier (see Note 1 below).

Note 1 An LVV Certifier appointed to certify under LVV certification category 'LV1B Modified Production – Extended' is also authorised to issue *LVVTA FS012 Upper Seatbelt Anchorage Request Forms* for vehicles which can be excluded from the requirement to be retro-fitted with upper seatbelt anchorages.

2.3 Category LV1C Modified Production – Structures

- 2.3(1) An LVV Certifier appointed to certify under LVV Certifier category 'LV1C Modified Production – Structures' (an LV1C Certifier) is authorised to certify all Class M-group and Class NA modified production low volume vehicles to the extent specified in 2.3 as **'may be certified'** (see Note 1 below).

Note 1 Class M-group means passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

Body structure

- 2.3(2) Body structure modifications and conversions which **may be certified** by an LV1C Certifier are, subject to 2.3(3):

- (a) all body modifications specified in 2.1(13); and
- (b) all structural body modifications and conversions, including restructuring and configuration changes, which retain the vehicle manufacturer's occupant protection systems and structure forward of the A-pillars; and
- (c) complete ('bolt-off/on' OEM-type) body substitution.

- 2.3(3) Body structure modifications and conversions which **must not be certified** by an LV1C Certifier are:

- (a) modifications and conversions which do not retain the vehicle manufacturer's occupant protection systems and structure forward of the A-pillars; or
- (b) substantial or complete custom body construction; or
- (c) those which are not listed in 2.3(2).

Chassis structure

- 2.3(4) Chassis structure modifications, conversions, and construction features which **may be certified** by an LV1C Certifier **are**, subject to 2.3(5):
- (a) modifications to OEM chassis rails or sub-frame rails rearward of the A-pillars; and
 - (b) modifications to, or substitution or construction of, cross-members; and
 - (c) modifications to body mounting outriggers for the purpose of providing tyre clearance.
- 2.3(5) Chassis structure modifications, conversions, and construction features which **must not be certified** by an LV1C Certifier **are**:
- (a) modifications made to OEM chassis rails or sub-frame rails forward of the A-pillars; or
 - (b) complete chassis substitution or construction; or
 - (c) complete rear half-chassis construction; or
 - (d) complete sub-frame changes; or
 - (e) those which are not listed in 2.3(4).

2.4 Category LV1D Advanced Modified Production & Scratch-built

- 2.4(1) An LVV Certifier appointed to certify under LVV certification category 'LV1D Advanced Modified Production & Scratch-built' (an LV1D Certifier) is authorised to certify all Class M-group and Class NA modified production and scratch-built low volume vehicles to the extent specified in 2.4 as **'may be certified'** (see Note 1 below).

Note 1 Class M-group means passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

- 2.4(2) Modifications, conversions, and construction features which **may be certified** by an LV1D Certifier **are**:
- (a) **those** modifications and conversions specified as 'may be certified' in 2.1, 2.2, and 2.3; and
 - (b) all engine and drive-train conversions; and
 - (c) all suspension modifications, conversions, and construction features, including the design and construction of custom-built suspension components and systems, the relocation or substitution of major non-OEM suspension components and systems, and changes which could affect the suspension geometry or performance; and
 - (d) all steering modifications, conversions, and construction features, including the design and construction of custom-built steering components and systems, the relocation or substitution of major non-OEM steering components and systems, and changes which could affect the steering geometry or performance; and
 - (e) all braking modifications, conversions, and construction features, including the design and construction of custom-built braking components and systems, the relocation or substitution of major non-OEM braking components and systems, and changes which could affect brake balance or performance; and
 - (f) all modifications, conversions, construction, and installations of seats and seat anchorages, and seatbelt anchorages (see Note 1 below); and

- (g) all body modifications, conversions, and construction features, including body restructuring, configuration changes, body type or style substitution, and complete body design and construction; and
- (h) all chassis modifications, conversions, and construction features, including the design and construction of custom-built chassis and chassis elements, the substitution of major chassis and chassis elements, and changes which could affect the vehicle's torsional and longitudinal rigidity; and
- (i) all modifications, conversions, and construction features to vehicles of Class M-group and Class NA which have been issued with an individual approval from the LVVTA Technical Advisory Committee (see Note 2 below); and
- (j) a low volume vehicle for which an LVV Authority Card issued by the New Zealand Hot Rod Association specifying 'Fender Exemption' has been issued (see Note 3 below).

Note 1	An LVV Certifier appointed to certify under LVV certification category 'LV1D Advanced Modified Production & Scratch-built' is also authorised to issue <i>LVVTA FS012 Upper Seatbelt Anchorage Request Forms</i> for vehicles which can be excluded from the requirement to be retro-fitted with upper seatbelt anchorages.
Note 2	<u>Only</u> an LV1D Certifier can LVV certify a Class M-group or Class NA low volume vehicle which features a component or system that has been issued with an individual approval from the LVVTA Technical Advisory Committee (TAC).
Note 3	<u>Only</u> an LV1D Certifier can LVV certify a low volume vehicle for which a New Zealand Hot Rod Association LVV Authority Card which specifies 'Fender Exemption' has been issued.

2.4(3) Modifications, conversions, and construction features which **must not be certified** by an LV1D Certifier **are** those not listed in 2.4(2).

Section 3 Category 2: Motorcycles & Trikes

3.1 Category LV2A Motorcycles - Modified Production

3.1(1) An LVV Certifier appointed to certify under LVV certification category 'LV2A Motorcycles - Modified Production' (an LV2A Certifier) is authorised to certify all Class LC modified production low volume motorcycles to the extent specified in 3.1 as '**may be certified**' (see Note 1 below).

Note 1	Class LC means two-wheeled motorcycles. A full list and descriptions of NZTA's Table A vehicle classes are provided in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> , which is available from NZTA's website https://www.nzta.govt.nz/
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Engine and drive-train

3.1(2) All engine and drive-train modifications and conversions **may be certified** by an LV2A Certifier.

Braking systems

3.1(3) Braking system modifications and conversions which **may be certified** by an LV2A Certifier **are**, subject to 3.1(4):

- (a) substitution and adaptation of 'bolt-on' OEM or aftermarket braking components and systems; and
- (b) modification and adaptation of hand and foot control systems; **and**
- (c) disabling or removal of OEM anti-lock braking systems; **and**

(d) conversion of a combined braking system to an individual braking system.

3.1(4) Braking system modifications and conversions which **must not be certified** by an LV2A Certifier are:

- (a) custom-built braking components or systems; or
- (b) those which are not listed in 3.1(3).

Steering systems

3.1(5) Steering system modifications and conversions which **may be certified** by an LV2A Certifier are, subject to 3.1(6), the substitution and adaptation of 'bolt-on' OEM or aftermarket steering components and systems.

3.1(6) Steering system modifications and conversions which **must not be certified** by an LV2A Certifier are:

- (a) custom-fabricated steering components or systems; or
- (b) complete steering head adaptations; or
- (c) those which are not listed in 3.1(5).

Suspension systems

3.1(7) Suspension system modifications and conversions which **may be certified** by an LV2A Certifier are, subject to 3.1(8):

- (a) a hard-tail conversion (see Note 1 below); and
- (b) the substitution or adaptation of 'bolt-on' OEM or aftermarket suspension components and systems.

Note 1 A conversion to a 'hard-tail' rear end on a motorcycle replaces the rear swing-arm and the rear suspension assembly, so the ability for an LV2A Certifier to LVV certify a hard-tail conversion is not subject to the requirements of 3.1(8)(a) or (b).

3.1(8) Suspension system modifications and conversions which **must not be certified** by an LV2A Certifier are:

- (a) custom-fabricated suspension components or systems; or
- (b) custom-fabricated rear swing-arms; or
- (c) those which are not listed in 3.1(7).

Wheels and tyres

3.1(9) All wheel and tyre modifications and changes **may be certified** by an LV2A Certifier.

Fuel systems

3.1(10) All fuel system modifications and conversions **may be certified** by an LV2A Certifier, including custom-built fuel tanks and systems.

Frame and bodywork

3.1(11) Frame and bodywork modifications and conversions which **may be certified** by an LV2A Certifier are, subject to 3.1(12):

- (a) bodywork modification and substitution; and
- (b) minor frame modifications (see Note 1 below).

Note 1 'Minor frame modifications' can include modifications to engine and drive-train mounting systems, seat mounts, and mounting systems for sidecar attachment.

3.1(12) Frame and bodywork modifications and conversions which **must not be certified** by an LV2A Certifier are:

- (a) custom-built frames; or
- (b) those which are not listed in 3.1(11).

Sidecars

3.1(13) All side-car modifications and conversions **may be certified** by an LV2A Certifier.

3.2 Category LV2B Motorcycles – Advanced Modified Production & Scratch-built

3.2(1) An LVV Certifier appointed to certify under LVV certification category 'LV2B Motorcycles – Advanced Modified Production & Scratch-built' (an LV2B Certifier) is authorised to certify all Class LC modified production and scratch-built low volume motorcycles to the extent specified in 3.2 as **may be certified** (see Note 1 below).

Note 1 Class LC means two-wheeled motorcycles. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

3.2(2) Modifications, conversions, and construction features which **may be certified** by an LV2B Certifier are:

- (a) all modifications and conversions specified as 'may be certified' in 3.1; and
- (b) all braking modifications, conversions, and construction features, including the design and construction of custom-built braking components and systems, the relocation or substitution of major non-OEM braking components and systems, and changes which could affect brake balance or performance; and
- (c) all steering modifications, conversions, and construction features, including the design and construction of custom-built steering components and systems (in particular steering head adaptations), the relocation or substitution of major non-OEM steering components and systems, and changes which could affect the steering geometry or performance; and
- (d) all suspension modifications, conversions, and construction features, including the design and construction of custom-built suspension components and systems (in particular front ends and swing-arm assemblies), the relocation or substitution of major non-OEM suspension components and systems, and changes which could affect the suspension geometry or performance; and
- (e) all body and frame modifications, conversions, and construction features, including the design and construction of custom-built frames and frame elements, the substitution of major frame and frame elements, and changes which could affect the motorcycle's torsional and longitudinal rigidity; and

- (f) all modifications, conversions, and construction features to vehicles of Class L-group (except for Class LE) which have been individually approved by the LVVTA Technical Advisory Committee (see Note 1 below).

Note 1 Only an LV2B Certifier can LVV certify a Class L-group (but not Class LE) low volume vehicle which features a component or system that has been issued with an individual approval from the LVVTA Technical Advisory Committee (TAC).

- 3.2(3) Modifications, conversions, and construction features which **must not be certified** by an LV2B Certifier **are** those not listed in 3.2(2).

3.3 Category LV2C Trikes – Advanced Modified Production & Scratch-built

- 3.3(1) An LVV Certifier appointed to certify under LVV certification category 'LV2C Trikes – Advanced Modified Production & Scratch-built' (an LV2C Certifier) is authorised to certify all Class LE and Class MA modified production and scratch-built low volume trikes to the extent specified in 3.2 as '**may be certified**' (see Notes 1 and 2 below).

Note 1 Class LE means a trike (motor-tricycle or three-wheeled motor vehicle). Some 'heavy' trikes cannot be certified as Class LE vehicles, and therefore must be treated as Class MA passenger vehicles (cars). A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

Note 2 Only an LV2C Certifier can LVV certify a Class LE or Class MA low volume trike which features a component or system that has been issued with an individual approval from the LVVTA Technical Advisory Committee (TAC).

- 3.3(2) Modifications, conversions, and construction features which **must not be certified** by an LV2C Certifier **are** those not referred to in 3.3(1).

Section 4 Category 3: Disability Adaptation & Transportation

4.1 Category LV3A Disability Adaptation – Limited

- 4.1(1) An LVV Certifier appointed to certify under LVV certification category 'LV3A Disability Adaptation - Limited' (an LV3A Certifier) is authorised to certify all Class M-group and Class NA modified production low volume vehicles to the extent specified in 4.1 as '**may be certified**' (see Note 1 below).

Note 1 **Class M-group** means passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

- 4.1(2) Modifications and conversions which **may be certified** by an LV3A Certifier **are**, subject to 4.1(3):

(a) modifications and conversions carried out to enable a person with a disability to:

(i) (together with a caregiver or attendant) enter or exit a vehicle; and

(ii) be seated and restrained within a vehicle;

and

(b) those adaptive control systems specified in 2.2(4).

- 4.1(3) Disability adaptation modifications and conversions which **must not be certified** by an LV3A Certifier **are** those not listed in 4.1(2).

4.2 Category LV3B Disability Adaptation – Structures

- 4.2(1) An LVV Certifier appointed to certify under LVV certification category 'LV3B Disability Adaptation – Structures (an LV3B Certifier) is authorised to certify Class M-group and Class NA modified production low volume vehicles to the extent specified in 4.2 as '**may be certified**' (see Note 1 below).

Note 1 Class M-group means passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

- 4.2(2) Modifications and conversions which **may be certified** by an LV3B Certifier are, subject to 4.2(3):
- (a) all modifications, conversions, and construction features specified as 'may be certified' in 4.1; and
 - (b) electro-mechanical driver control systems; and
 - (c) advanced steering, braking, or other driver control systems which feature an unusually high level of complexity, such as those that do not incorporate a mechanical connection; and
 - (d) modifications carried out for the purpose of disability transportation that affect the integrity and rigidity of the permanent vehicle structure, including restructuring and configuration changes rearward of the A-pillars.
- 4.2(3) Structural modifications and conversions which **must not be certified** by an LV3B Certifier are:
- (a) those which affect the vehicle manufacturer's occupant protection systems and structure forward of the A-pillars; or
 - (b) those which are not listed in 4.2(2).

Section 5 Category 4: Electric & Hybrid Vehicles

5.1 Category LV4 - Electric & Hybrid Vehicles

- 5.1(1) An LVV Certifier appointed to certify under LVV certification category 'LV4 – Electric & Hybrid Vehicles' (an LV4 Certifier) is authorised to certify Class M-group, Class NA, and Class L-group modified production and scratch-built low volume vehicles which are converted to, or constructed, using electric or hybrid motive power, to the extent specified in 5.1 as '**may be certified**' (see Note 1 below).

Note 1 Class M-group means passenger-carrying cars, station wagons, and vans, Class NA means light goods-carrying utilities and vans, and Class L-group means motorcycles and trikes. A full list and descriptions of NZTA's Table A vehicle classes are provided in *Land Transport Rule: Vehicle Standards Compliance 2002*, which is available from NZTA's website <https://www.nzta.govt.nz/>

- 5.1(2) Modifications and conversions which **may be certified** by an LV4 Certifier are, subject to 5.1(3):
- (a) the electrical aspects of the modified, converted, or constructed vehicle; and
 - (b) the battery box design and attachment; and
 - (c) a vacuum pump system provided to aid braking; and
 - (d) an electric power steering pump.

- 5.1(3) Modifications and conversions which **must not be certified** by an LV4 Certifier are:
- (a) any engine mounting systems; or
 - (b) any mechanical and engineering aspects of the modified, converted, or constructed vehicle which are not listed in 5.1(2) (see Notes 1 to 3 below).

Note 1	An LV4 Certifier is authorised to LVV certify only those aspects of a modified or scratch-built low volume vehicle which relate to the electric vehicle or hybrid conversion or construction features. All mechanical and engineering-related aspects of an electric vehicle conversion (or construction) must be LVV certified by an LV1A or LV1D Certifier, as appropriate.
Note 2	There may be some cases where an LVV Certifier who holds both LV1A (or LV1D), and LV4, is not available within a geographical region. In this case, if a vehicle owner doesn't want to take the electric vehicle to a region where an LVV Certifier who holds both LV1A (or LV1D), and LV4, is available, two LVV Certifiers (one with LV1A or LV1D, and one with LV4) may need to be engaged to jointly certify the electric vehicle.
Note 3	The reason that the situation described in Note 2 above may exist is because very few LV4 Certifiers are appointed in New Zealand. This is because there is a high level of expertise required to LVV certify electric vehicles, but very little LVV certification activity happening - so it is not economic to have several LV4 Certifiers around the country all doing very little electric vehicle LVV certification work.

Section 6 Category 5: RHD Conversion, Exhaust, & Authority Cards

6.1 Category LVRH Modified Production – Right-hand Drive Conversions

- 6.1(1) An LVV Certifier appointed to certify under LVV certification category 'LVRH Modified Production – Right-hand Drive Conversions' (an LVRH Certifier) is authorised to certify all Class M-group and Class NA modified production low volume vehicles which have been converted from left-hand drive to right-hand drive, provided that the conversion is of a direct mirror-image nature (see Note 1 below).

Note 1	Class M-group means passenger-carrying cars, station wagons, and vans, and Class NA means light goods-carrying utilities and vans. A full list and descriptions of NZTA's Table A vehicle classes are provided in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> , which is available from NZTA's website https://www.nzta.govt.nz/
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6.2 Category LVEX – Exhaust Noise & Gas Emissions

- 6.2(1) An LVV Certifier appointed to certify under LVV certification category 'LVEX – Exhaust Noise & Gas Emissions' (an LVEX Certifier) is authorised to certify all Class M-group, Class NA, and Class L-group modified production and scratch-built low volume vehicles which require the testing of (see Note 1 below):
- (a) exhaust noise emissions; or
 - (b) exhaust gas emissions.

Note 1	Class M-group means passenger-carrying cars, station wagons, and vans, Class NA means light goods-carrying utilities and vans, and Class L-group means motorcycles and trikes. A full list and descriptions of NZTA's Table A vehicle classes are provided in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> , which is available from NZTA's website https://www.nzta.govt.nz/
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6.3 Category LVAC - Authority Card

- 6.3(1) An LVV Certifier appointed to certify under LVV certification category 'LVAC - Authority Card' (an LVAC Certifier) is authorised to certify Class M-group and Class NA modified production low volume vehicles which must meet alternative occupant protection equipment requirements for sporting purposes, as provided for in 2.12 of the *Low Volume Vehicle Code* (see Note 1 below).

Note 1 The *Low Volume Vehicle Code*, as referred to in 6.3(1), is available to the public electronically, free of charge, from the LVVTA website www.lvvta.org.nz

Section 7 Associated Information

7.1 Category Extensions

7.1(1) On a case-by-case basis, an LVV Certifier may certify a vehicle which is outside of the specified LVV certification category for which they are appointed, via a pre-approval process from LVVTA called a 'Category Extension' (see Note 1 below).

Note 1 LVV 'Category Extensions', as referred to in 7.1(1), are detailed in *LVV ORS Chapter 8: LVV Certification Inspection Procedures*.

Terms & Definitions for Chapter 3

A-arm	means a set of suspension control arms which incorporate two pivot points on the inner end and one pivot point on the outer end, usually found in an independent front suspension.
Adaptive control systems	means controls for people who are prevented, by a physical disability, from driving a vehicle using conventional controls, in particular those primary controls that operate the brake and accelerator systems.
Aftermarket	means a component or system made by a manufacturer, other than a high-volume motor vehicle manufacturer, who produces catalogued components or systems on a production-run basis for the mass-market.
A-Pillar	means the forward-most pillars in a vehicle structure which support the windscreen and the front of the roof, and from which the front doors are usually hinged.
Bump-stop	means a compression spring, usually of rubber, that prevents contact, or reduces the severity of contact, between the vehicle structure and a suspension component, when the vehicle's suspension system encounters a bump that causes the suspension to fully compress.
Butt-joint	means a method of attaching two components together by welding, where there is no overlap between the surfaces and the weld is run along the edge of the component.
Caster	means the angle between the steering axis and the vertical axis of a vehicle, when viewed in side elevation.
Certify	is, as defined in the <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> , to verify that a vehicle complies with safety-related legal requirements prescribed by New Zealand land transport legislation.
Chassis	(also known as a frame, or chassis frame) means a vehicle's supporting frame or platform to which the major mechanical components and body attach.

Cross-member	means a section or beam which connects two parallel sections of a chassis or sub-frame, sometimes incorporating a load-bearing mount for engine, drive-train, and steering and suspension components.
Custom	means a component or system fabricated by an individual person or small company on a one-off or limited-run basis, which is not intended as a high volume catalogued aftermarket part.
Differential	means the mechanical assembly used for transferring the engine power output, via the drive-train, to the driving wheels.
Disability adaptation	means modifications made to a vehicle to accommodate the specific needs of a person with a disability, particularly in relation to entering, being secured in, and exiting a vehicle.
Drive-shaft	means the assembly which transfers the power output from the gearbox to the differential.
Drive-train	means the components of a vehicle which transfer the engine power output to the differential, including the gearbox, flywheel, clutch, drive-shaft, and universal joints.
Drop-box	means a bolt-on component used to lower the chassis-end radius arm mounts on a solid-axle four-wheel drive vehicle to improve suspension geometry.
Electric vehicle	means a vehicle which has a form of motive power derived from electricity, rather than internal combustion.
External projections	means parts of a vehicle which protrude beyond the body surfaces that could present a hazard to pedestrians, including functional items such as bull bars, and non-functional items such as hood ornaments.
Fender exemption	(also known as a mudguard exemption) means a concession enabled by the issue of an Lvv Authority Card for owners of certain vehicle types to operate a vehicle without fenders.
Frontal impact protection	means the systems built into a vehicle, which include supplementary restraint system airbags, which are designed to prevent or reduce injury to the occupants in the event of a frontal collision.
Gearbox	means the mechanical assembly used to convert engine speed to road speed through the use of a number of different gear ratios.
Gearbox tunnel	means the floor of a vehicle surrounding the area where the gearbox is located.
Hard-tail	means, on a motorcycle or trike, an assembly which solidly connects the rear suspension arms to the frame, with the associated removal of the rear suspension system.
HICAS	(High Capacity Actively Controlled Steering) is a rear-wheel (auxiliary) steering system which was fitted to some Nissan vehicle models between approximately 1985 to 2010.
Hybrid vehicle	means a vehicle which uses more than one type of propulsion system, typically combining an internal combustion engine with one or more electric motors.
ICE	(Internal Combustion Engine) means an engine which generates motive power as the direct result of the explosive force of the combustion of a fuel and air mixture.

Inspection	means the vehicle inspection process specified in <i>section 2.4, 2.5, and 2.6 of the LVV Code</i> , carried out by an LVV Certifier during the LVV certification of a low volume vehicle.
L-class	is an NZTA vehicle classification, which means, in simple terms, two or three-wheeled motor vehicles which have a GVM of less than 1000 kg. More detailed information about L-class vehicles can be found in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> .
Low Volume Motorcycle	means, in simple terms, motorcycles which are modified or scratch-built in small numbers, and includes individually modified or scratch-built motorcycles. (See also the definition for an LVV).
LVV	(Low Volume Vehicle) means, in simple terms, vehicles which are modified or scratch-built in small numbers, and includes individually modified or scratch-built vehicles. The full definition of an LVV is contained in the <i>LVV Code</i> .
LVV Authority Card	(Low Volume Vehicle Authority Card) means an LVV certification document, issued under the delegated authority of LVVTA, specifying alternative safety related equipment required to be fitted to a vehicle for special purposes, as defined in <i>Annex 5 of the LVV Code</i> .
LVV Certification	(Low Volume Vehicle Certification) means the process specified by the <i>LVV Code</i> , by which the design of an LVV is determined to comply with any applicable requirements, and, in recognition of which, an LVV EDP is affixed.
LVV Certifier	(Low Volume Vehicle Certifier) means a person appointed by NZTA under the provisions of <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> , to carry out certification of modified and scratch-built LVVs, as specified by <i>Part 2 of the LVV Code</i> .
LVV Code	(Low Volume Vehicle Code or the Code) means an LVVTA document which is incorporated by reference into the <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> , and all applicable individual <i>Land Transport equipment rules</i> , that provides the legal framework to enable the LVV certification of modified and scratch-built LVVs in New Zealand.
LVV Information Sheets	(Low Volume Vehicle Information Sheets) means <i>Information Sheets</i> incorporated by reference under the <i>LVV Code</i> , which provide or support applicable requirements.
LVV Supporting Documents	(Low Volume Vehicle Supporting Documents) means those documents listed in <i>Annex 4 of the LVV Code</i> which comprise part of the applicable requirements, and support the application of the LVV certification system.
LVVTA	(Low Volume Vehicle Technical Association) is an incorporated society comprised of specialist vehicle associations. Established in 1992, its objectives are to represent the interests of vehicle modifiers and builders in New Zealand, and to ensure high safety standards for modified and scratch-built LVVs. The LVVTA owns and administers the <i>LVV Code</i> .
LVVTA TAC	(LVVTA Technical Advisory Committee) is an LVVTA-appointed panel of industry expert-level technical specialists, established to provide LVVTA with a very high level of technical support and direction on all technical matters relevant to the LVV certification system.

Macpherson strut	means a telescopic independently sprung suspension member incorporating a shock absorber, which is fixed at its upper end to the body shell or chassis, and has the lower end located by linkages which control transverse and fore and aft movement.
Mass-produced vehicle	(also known as production vehicle, or high-volume vehicle) means a vehicle which is manufactured in quantities of more than 500 at any one location in any one year for the mass market.
Modification	is defined in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> to change a vehicle from its original state by altering, substituting, adding or removing any structure, system, component or equipment, but does not include repair. 'Modified' and 'modification' have corresponding meanings.
Modified Production (LVV)	means, in simple terms, a vehicle which, while modified, maintains a sufficient percentage of body or chassis from one primary mass-produced vehicle that it can still be considered to be that vehicle. The full legal definition of a Modified Production LVV is complex and currently under review, and will be incorporated within the <i>LVV Code</i> once revised.
Motor vehicle	is as defined in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> .
NZHRA	(New Zealand Hot Rod Association) is an organisation which administers hot rodding within New Zealand, and is a founding member of the LVVTA.
NZTA	(New Zealand Transport Agency is a Crown entity responsible for managing New Zealand's land transport system.
Occupant protection system	means the systems built into a vehicle that are designed to prevent injury to occupants in the event of a collision, which can include energy-absorbing structural sections, front and side curtain airbags, and pretensioner seatbelts.
OE	(Original Equipment) means the parts and equipment used in the assembly process of a mass-produced vehicle.
OEM	(Original Equipment Manufacturer) means a company that produces the parts and equipment used in the assembly process of a mass-produced vehicle.
ONT	(Objective Noise Test) means a test carried out by an LVV Certifier to measure the noise level of a vehicle's exhaust system at a specific engine RPM.
Pitch Circle Diameter	(PCD) means the arrangement of a vehicle's wheel-mounting studs or bolts around an imaginary circle. For example, a vehicle with a PCD of 5 x 114 would have 5 studs arranged equidistantly around a circle of 114 mm diameter.
Retro-fit	means the fitting of a component or system that was not originally provided as original equipment during the assembly of the vehicle.
Right-hand drive	means that a vehicle's steering wheel and driving position is located to the right of the vehicle's longitudinal centreline.
Scratch-built (LVV)	means, in simple terms, an LVV which has been individually constructed from unrelated components, or a mass-produced vehicle which has been modified to such an extent that it can no longer be considered to be a modified mass-produced vehicle. The full legal definition of a scratch-built LVV is currently under review, and will be incorporated within the <i>LVV Code</i> once revised.

Steering box	means a device which converts the rotary motion of the steering wheel and shaft into the linear motion of a steering arm connected to a parallelogram-like steering linkage.
Stressed seat	means a seat that has one or more seatbelt anchorages attached to the seat frame in such a way that any loading applied to the seatbelt anchorage would be transmitted through the seat structure to the seat anchorage.
Structure	means the elements of a vehicle which are critical to the strength, rigidity, steering and suspension geometry, and crashworthiness of the vehicle.
Stub axle spindle	(also known as a pin or axle) means the machined section of a stub axle which the hub bearings within a suspension assembly run on.
Sub-frame	means a structural element of a unitary-construction vehicle to which the major mechanical components, including the engine, drive-train, and suspension, attach.
Suspension cross-member	means a section or beam which connects two parallel sections of the main chassis or sub-frame, and is used to support a suspension system.
Table A vehicle classes	means NZTA's classification of vehicle types, as defined in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> .
Trike	means, usually, a Class LE motor-tricycle (three-wheeled motor vehicle), as defined in <i>Land Transport Rule: Vehicle Standards Compliance 2002</i> .
Unitary-construction	(also known as unitary-constructed) means a type of vehicle construction which incorporates the vehicle body and chassis frame in one unit, rather than having a separate and removable chassis.
Unstressed seat	means a seat that does not have any seatbelt anchorages attached to the seat frame.
Wishbone arm	means the same as A-arm.